Sequences

CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

August 19, 2026

Department of CSE, IIT Hyderabad

Sequences

Some Sequences

Key Idea

A *sequence* associates an element with each integer index.

- An array can be viewed as a finite sequence of values.
- A string is a sequence of characters.
- The steps of an algorithm can be represented by a sequence of states.
- Measurements collected over time form a sequence.

Some Sequences

Key Idea

A *sequence* associates an element with each integer index.

- An array can be viewed as a finite sequence of values.
- A string is a sequence of characters.
- The steps of an algorithm can be represented by a sequence of states.
- Measurements collected over time form a sequence.

Definition of a Sequence

Definition

A **sequence** is a function from a subset of integers (usually starting with $\{0, 1, 2, \ldots\}$ or $\{1, 2, 3, \ldots\}$) to a set S.

- We use the notation a_n to represent the image of the integer n.
- a_n is called a **term** of the sequence.
- $\{a_n\}$ represents the sequence itself.

Example: Let $a_n = 1/n$ for $n \in \mathbb{Z}^+$. The sequence is:

$$\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$$

A Sequence Is a Function

A sequence is simply a function whose inputs are restricted to integer indices.

Ordinary Function		**Sequence**
	$\longrightarrow$	
$f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2$		$a : \mathbb{Z}^+ \to \mathbb{R},\ a_n = n^2$
The input can be any real number:		The input is an integer index:
$x = 2.5,\ -1,\ \sqrt{2}, \ldots$		$n = 1, 2, 3, \ldots$

Example

For $a_n = n^2$, $a_1 = 1, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 16, \ldots$
Thus the sequence is

$$\boxed{1, 4, 9, 16, 25, \ldots}$$

A Sequence Is a Function

A sequence is simply a function whose inputs are restricted to integer indices.

Ordinary Function		**Sequence**
	$\longrightarrow$	
$f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2$		$a : \mathbb{Z}^+ \to \mathbb{R},\ a_n = n^2$
The input can be any real number:		The input is an integer index:
$x = 2.5,\ -1,\ \sqrt{2}, \ldots$		$n = 1, 2, 3, \ldots$

Example

For $a_n = n^2$, $a_1 = 1, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 16, \ldots$

Thus the sequence is

$$\boxed{1, 4, 9, 16, 25, \ldots}$$

Can You Spot the Pattern?

$\mathcal{Q}$. For each sequence, determine the next three terms.

1. $5,\ 8,\ 11,\ 14,\ \ldots$
2. $2,\ 6,\ 18,\ 54,\ \ldots$
3. $100,\ 50,\ 25,\ 12.5,\ \ldots$

Question: What rule generates each sequence?

Can You Spot the Pattern?

$\mathcal{Q}$. For each sequence, determine the next three terms.

1. $5,\ 8,\ 11,\ 14,\ \ldots$
2. $2,\ 6,\ 18,\ 54,\ \ldots$
3. $100,\ 50,\ 25,\ 12.5,\ \ldots$

Question: What rule generates each sequence?

$\mathcal{A}$. An *arithmetic progression* is obtained by adding the **same number** repeatedly.

Arithmetic Progressions (AP)

Definition

An **arithmetic progression** is a sequence of the form:

$$a, a+d, a+2d, a+3d, \ldots, a+nd, \ldots$$

where the initial term a and the common difference d are real numbers.

ex. Let $a = -1$, $d = 4$: $\{-1, 3, 7, 11, 15, \ldots\}$

ex. Let $a = 7$, $d = -3$: $\{7, 4, 1, -2, -5, \ldots\}$

ex. Let $a = 1$, $d = 2$: $\{1, 3, 5, 7, 9, \ldots\}$

AP can be viewed as a function of its index: $f(n) = a_n = a + (n-1)d$.

Can You Spot the Pattern?

$\mathcal{Q}$. For each sequence, determine the next three terms.

- ▷ $3, 6, 12, 24, \ldots$
- ▷ $81, 27, 9, 3, \ldots$
- ▷ $5, 10, 20, 40, \ldots$
- ▷ $64, 32, 16, 8, \ldots$

Question: What rule generates each sequence?

Can You Spot the Pattern?

$\mathcal{Q}$. For each sequence, determine the next three terms.

- ▷ $3,\ 6,\ 12,\ 24,\ \ldots$
- ▷ $81,\ 27,\ 9,\ 3,\ \ldots$
- ▷ $5,\ 10,\ 20,\ 40,\ \ldots$
- ▷ $64,\ 32,\ 16,\ 8,\ \ldots$

Question: What rule generates each sequence?

$\mathcal{A}$. A geometric progression is obtained by multiplying by the **same number** repeatedly.

Geometric Progressions (GP)

Definition

A **geometric progression** is a sequence of the form:

$$a, ar, ar^2, ar^3, \ldots, ar^n, \ldots$$

where the initial term a and the common ratio r are real numbers.

ex. Let $a = 1$, $r = -1$: $\{1, -1, 1, -1, 1, \ldots\}$

ex. Let $a = 2$, $r = 5$: $\{2, 10, 50, 250, 1250, \ldots\}$

ex. Let $a = 6$, $r = 1/3$: $\{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots\}$

GP can be viewed as a function of its index: $f(n) = a_n = ar^{n-1}$.

Geometric Progressions (GP)

Definition

A **geometric progression** is a sequence of the form:

$$a, ar, ar^2, ar^3, \ldots, ar^n, \ldots$$

where the initial term a and the common ratio r are real numbers.

ex. Let $a = 1$, $r = -1$: $\{1, -1, 1, -1, 1, \ldots\}$

ex. Let $a = 2$, $r = 5$: $\{2, 10, 50, 250, 1250, \ldots\}$

ex. Let $a = 6$, $r = 1/3$: $\{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots\}$

GP can be viewed as a function of its index: $f(n) = a_n = ar^{n-1}$.

Arithmetic vs. Geometric Progressions

	Arithmetic (AP)	Geometric (GP)
Next term	$a_{n+1} = a_n + d$	$a_{n+1} = a_n \cdot r$
Constant	Difference d	Ratio r
nth term	$a_n = a + (n-1)d$	$a_n = ar^{n-1}$

AP: Repeated Addition

$2,\ 5,\ 8,\ 11,\ 14, \ldots$

$+3 \quad +3 \quad +3 \quad +3$

GP: Repeated Multiplication

$2,\ 6,\ 18,\ 54,\ 162, \ldots$

$\times 3 \quad \times 3 \quad \times 3 \quad \times 3$

Arithmetic vs. Geometric Progressions

	Arithmetic (AP)	Geometric (GP)
Next term	$a_{n+1} = a_n + d$	$a_{n+1} = a_n \cdot r$
Constant	Difference d	Ratio r
nth term	$a_n = a + (n-1)d$	$a_n = ar^{n-1}$

AP: Repeated Addition

$$2,\ 5,\ 8,\ 11,\ 14, \ldots$$

$$+3 \quad +3 \quad +3 \quad +3$$

GP: Repeated Multiplication

$$2,\ 6,\ 18,\ 54,\ 162, \ldots$$

$$\times 3 \quad \times 3 \quad \times 3 \quad \times 3$$

A Sequence Puzzle

$\mathcal{Q}$. Consider the following sequence: $2,\ 5,\ 8,\ 14,\ 23,\ 38,\ \ldots$

1. Can you guess the rule?
2. What is the next term?

A Sequence Puzzle

$\mathcal{Q}$. Consider the following sequence: $2,\ 5,\ 8,\ 14,\ 23,\ 38,\ \ldots$

1. Can you guess the rule?
2. What is the next term?

Hint: Look only at two previous terms.

A Sequence Puzzle

$\mathcal{Q}$. Consider the following sequence: $2,\ 5,\ 8,\ 14,\ 23,\ 38,\ \ldots$

1. Can you guess the rule?
2. What is the next term?

Hint: Look only at two previous terms.

$\mathcal{A}$. A *recurrence relation* is an equation that defines each term of a sequence in terms of one or more preceding terms.

Recurrence Relations

Definition

A **recurrence relation** for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms in the sequence $(a_0, a_1, \ldots, a_{n-1})$ for all integers n with $n \geq n_0$.

- Let $a_n = a_{n-1} + 3$ for $n \geq 1$. Suppose the initial condition is $a_0 = 2$.

$$a_1 = a_0 + 3 = 2 + 3 = 5$$
$$a_2 = a_1 + 3 = 5 + 3 = 8$$
$$a_3 = a_2 + 3 = 8 + 3 = 11$$

Recurrence Relations

Definition

A **recurrence relation** for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms in the sequence $(a_0, a_1, \ldots, a_{n-1})$ for all integers n with $n \geq n_0$.

- Let $a_n = a_{n-1} + 3$ for $n \geq 1$. Suppose the initial condition is $a_0 = 2$.

$$
\begin{aligned}
a_1 &= a_0 + 3 = 2 + 3 = 5 \\
a_2 &= a_1 + 3 = 5 + 3 = 8 \\
a_3 &= a_2 + 3 = 8 + 3 = 11
\end{aligned}
$$

Examples

A company keeps track of its workforce. Suppose the number of employees this month depends on

- the number of employees last month, and
- half of the net increase from two months ago.

If H_n denotes the number of employees after month n, then

$$\boxed{H_n = H_{n-1} + \frac{1}{2}\left(H_{n-1} - H_{n-2}\right)}$$

Examples

A company keeps track of its workforce. Suppose the number of employees this month depends on

- the number of employees last month, and
- half of the net increase from two months ago.
- Can you compute the number of employees at the end of first year?

If H_n denotes the number of employees after month n, then

$$\boxed{H_n = H_{n-1} + \frac{1}{2}(H_{n-1} - H_{n-2})}$$

Examples

A company keeps track of its workforce. Suppose the number of employees this month depends on

- the number of employees last month, and
- half of the net increase from two months ago.
- Can you compute the number of employees at the end of first year?
- Initial condition: $H_0 = 100$ and $H_1 = 200$.

Examples

A company keeps track of its workforce. Suppose the number of employees this month depends on

- the number of employees last month, and
- half of the net increase from two months ago.
- Can you compute the number of employees at the end of first year?
- Initial condition: $H_0 = 100$ and $H_1 = 200$.

If H_n denotes the number of employees after month n, then

$$\boxed{H_n = H_{n-1} + \frac{1}{2}(H_{n-1} - H_{n-2})}$$

Example

A staircase has n steps. You can climb either

- one step at a time, or
- two steps at a time.

Let $F(n)$ denote the number of different ways to reach the n^{th} step.

n	1	2	3	4	5	6
$F(n)$	1	2	?	?	?	?

Question: How can we compute $F(3)$ without listing every possibility?

Example

A staircase has n steps. You can climb either

- one step at a time, or
- two steps at a time.

Let $F(n)$ denote the number of different ways to reach the n^{th} step.

n	1	2	3	4	5	6
$F(n)$	1	2	?	?	?	?

Question: How can we compute $F(3)$ without listing every possibility?

Notice: The answer for the current problem depends on answers to smaller problems!

Using Recurrence Relations

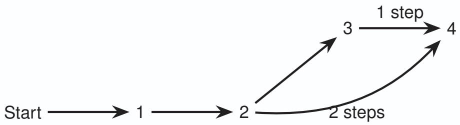
To reach the n^{th} step,

- the last move could be from step $(n-1)$, or
- the last move could be from step $(n-2)$.
- Therefore, $\boxed{F(n) = F(n-1) + F(n-2)}$, with $F(1) = 1$, $F(2) = 2$.

Using Recurrence Relations

To reach the n^{th} step,

- the last move could be from step $(n-1)$, or
- the last move could be from step $(n-2)$.
- Therefore, $\boxed{F(n) = F(n-1) + F(n-2)}$, with $F(1) = 1, F(2) = 2$.



- $F(2) = 2$ as $\{11, 2\}$
- $F(3) = 3$ as $\{111, 12, 21\}$
- $F(4) = 5 = F(3) + F(2)$ as $\{1111, 121, 211, 112, 22\}$

Example: The Fibonacci Sequence

Definition

The **Fibonacci sequence** $f_0, f_1, f_2, \ldots$ is defined by:

- **Initial Conditions:** $f_0 = 0$, $f_1 = 1$
- **Recurrence Relation:** $f_n = f_{n-1} + f_{n-2}$

$\mathcal{Q}$. Find f_2, f_3, f_4, f_5, f_6.

$\mathcal{A}$. **Solution:**

- $f_2 = f_1 + f_0 = 1 + 0 = 1$
- $f_3 = f_2 + f_1 = 1 + 1 = 2$
- $f_4 = f_3 + f_2 = 2 + 1 = 3$
- $f_5 = f_4 + f_3 = 3 + 2 = 5$
- $f_6 = f_5 + f_4 = 5 + 3 = 8$

Example: The Fibonacci Sequence

Definition

The **Fibonacci sequence** $f_0, f_1, f_2, \ldots$ is defined by:

- **Initial Conditions:** $f_0 = 0$, $f_1 = 1$
- **Recurrence Relation:** $f_n = f_{n-1} + f_{n-2}$

$\mathcal{Q}$. Find f_2, f_3, f_4, f_5, f_6.

$\mathcal{A}$. **Solution:**

- $f_2 = f_1 + f_0 = 1 + 0 = 1$
- $f_3 = f_2 + f_1 = 1 + 1 = 2$
- $f_4 = f_3 + f_2 = 2 + 1 = 3$
- $f_5 = f_4 + f_3 = 3 + 2 = 5$
- $f_6 = f_5 + f_4 = 5 + 3 = 8$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first five terms of the sequence $\{a_n\}$, where

$$a_n = 2(-3)^n + 5n, \qquad n = 0, 1, 2, \ldots$$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first five terms of the sequence $\{a_n\}$, where

$$a_n = 2(-3)^n + 5n, \qquad n = 0, 1, 2, \ldots$$

$\mathcal{A}$. Substitute $n = 0, 1, 2, 3, 4$:

$$\begin{aligned}
a_0 &= 2(-3)^0 + 5(0) = 2, \\
a_1 &= 2(-3)^1 + 5(1) = -1, \\
a_2 &= 2(-3)^2 + 5(2) = 28, \\
a_3 &= 2(-3)^3 + 5(3) = -39, \\
a_4 &= 2(-3)^4 + 5(4) = 182.
\end{aligned}$$

Therefore, the sequence begins

$$\boxed{\{2, -1, 28, -39, 202, \ldots\}}.$$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$, where

$$a_n = 1 + (-1)^n, \qquad n = 0, 1, 2, \ldots$$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$, where

$$a_n = 1 + (-1)^n, \qquad n = 0, 1, 2, \ldots$$

$\mathcal{A}$. Substitute $n = 0, 1, 2, 3, 4, 5$:

$$\begin{aligned} a_0 &= 1 + (-1)^0 = 2, \\ a_1 &= 1 + (-1)^1 = 0, \\ a_2 &= 1 + (-1)^2 = 2, \\ a_3 &= 1 + (-1)^3 = 0, \\ a_4 &= 1 + (-1)^4 = 2, \\ a_5 &= 1 + (-1)^5 = 0. \end{aligned}$$

Therefore,

$$\boxed{\{2, 0, 2, 0, 2, 0, \ldots\}}.$$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first five terms of the sequence $\{a_n\}$, where

$$a_n = (n+1)^{n+1}, \qquad n = 0, 1, 2, \ldots$$

Quick Check: Finding Terms of a Sequence

$\mathcal{Q}$. Find the first five terms of the sequence $\{a_n\}$, where

$$a_n = (n+1)^{n+1}, \qquad n = 0, 1, 2, \ldots$$

$\mathcal{A}$. Substitute $n = 0, 1, 2, 3, 4$:

$$\begin{aligned}
a_0 &= (0+1)^{0+1} = 1, \\
a_1 &= (1+1)^{1+1} = 2^2 = 4, \\
a_2 &= (2+1)^{2+1} = 3^3 = 27, \\
a_3 &= (3+1)^{3+1} = 4^4 = 256, \\
a_4 &= (4+1)^{4+1} = 5^5 = 3125.
\end{aligned}$$

Therefore,

$$\boxed{\{1, 4, 27, 256, 3125, \ldots\}}.$$

Quick Check: Describing a Sequence

$\mathcal{Q}$. Find an explicit formula for the sequence that lists the positive odd integers in increasing order, with each odd integer appearing twice.

Quick Check: Describing a Sequence

$\mathcal{Q}$. Find an explicit formula for the sequence that lists the positive odd integers in increasing order, with each odd integer appearing twice.

$\mathcal{A}$. The sequence begins

$$\{1, 1, 3, 3, 5, 5, 7, 7, 9, 9, \ldots\}.$$

The odd integer appearing in positions $2k - 1$ and $2k$ is $2k - 1$. Therefore, separating odd and even indices,

$$\boxed{a_n = \begin{cases} n, & n \text{ odd}, \\ n - 1, & n \text{ even}. \end{cases}}$$

Quick Check: A Sequence from Binary Expansions

$\mathcal{Q}$. Find the first ten terms of the sequence $\{a_n\}$, where a_n is the number of bits in the binary expansion of n.

Quick Check: A Sequence from Binary Expansions

$\mathcal{Q}$. Find the first ten terms of the sequence $\{a_n\}$, where a_n is the number of bits in the binary expansion of n.

$\mathcal{A}$. Write the integers in binary:

n	1	2	3	4	5	6	7	8	9	10
Binary	1	10	11	100	101	110	111	1000	1001	1010
a_n	1	2	2	3	3	3	3	4	4	4

Therefore,

$$\boxed{\{1, 2, 2, 3, 3, 3, 3, 4, 4, 4, \ldots\}}.$$

Quick Check: A Sequence from English Words

$\mathcal{Q}$. Find the first ten terms of the sequence $\{a_n\}$, where a_n is the number of letters in the English word for the integer n.

Quick Check: A Sequence from English Words

$\mathcal{Q}$. Find the first ten terms of the sequence $\{a_n\}$, where a_n is the number of letters in the English word for the integer n.

$\mathcal{A}$. The first ten positive integers are:

n	1	2	3	4	5	6	7	8	9	10
Word	one	two	three	four	five	six	seven	eight	nine	ten
a_n	3	3	5	4	4	3	5	5	4	3

Therefore,

$$\boxed{\{3, 3, 5, 4, 4, 3, 5, 5, 4, 3, \ldots\}}.$$

Quick Check: A Recursively Defined Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$ defined by

$$a_n = -2a_{n-1}, \qquad a_0 = -1.$$

Quick Check: A Recursively Defined Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$ defined by

$$a_n = -2a_{n-1}, \qquad a_0 = -1.$$

$\mathcal{A}$. Starting with $a_0 = -1$:

$$\begin{aligned}
a_1 &= -2a_0 = -2(-1) = 2, \\
a_2 &= -2a_1 = -2(2) = -4, \\
a_3 &= -2a_2 = -2(-4) = 8, \\
a_4 &= -2a_3 = -2(8) = -16, \\
a_5 &= -2a_4 = -2(-16) = 32.
\end{aligned}$$

Therefore,

$$\boxed{\{-1, 2, -4, 8, -16, 32, \ldots\}}.$$

Quick Check: A Recursively Defined Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$ defined by

$$a_n = -2a_{n-1}, \qquad a_0 = -1.$$

$\mathcal{A}$. Starting with $a_0 = -1$:

$$\begin{aligned}
a_1 &= -2a_0 = -2(-1) = 2, \\
a_2 &= -2a_1 = -2(2) = -4, \\
a_3 &= -2a_2 = -2(-4) = 8, \\
a_4 &= -2a_3 = -2(8) = -16, \\
a_5 &= -2a_4 = -2(-16) = 32.
\end{aligned}$$

Therefore,

$$\boxed{\{-1, 2, -4, 8, -16, 32, \ldots\}}.$$

$\mathcal{Q}$. What type of sequence is this?

Quick Check: A Recursively Defined Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$ defined by

$$a_n = na_{n-1} + n^2 a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

Quick Check: A Recursively Defined Sequence

$\mathcal{Q}$. Find the first six terms of the sequence $\{a_n\}$ defined by

$$a_n = na_{n-1} + n^2 a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

$\mathcal{A}$. Starting with $a_0 = 1$ and $a_1 = 1$:

$$\begin{aligned}
a_2 &= 2a_1 + 2^2 a_0 = 2(1) + 4(1) = 6, \\
a_3 &= 3a_2 + 3^2 a_1 = 3(6) + 9(1) = 27, \\
a_4 &= 4a_3 + 4^2 a_2 = 4(27) + 16(6) = 204, \\
a_5 &= 5a_4 + 5^2 a_3 = 5(204) + 25(27) = 1085.
\end{aligned}$$

Therefore, the sequence begins

$$\boxed{\{1, 1, 6, 27, 204, 1085, \ldots\}}.$$

Solving Recurrence Relations

A recurrence relation defines a sequence **indirectly**: each term is defined using previous terms

> Finding an explicit formula for the n-th term of a sequence generated by a recurrence relation is called **solving the recurrence relation.**

- This explicit formula is called a **closed-form formula.**
- We can solve these iteratively through *forward substitution* or *backward substitution*.

Iterative Solution: Forward Substitution

$\mathcal{Q}$. Let $a_n = a_{n-1} + 3$ for $n \geq 2$, with initial condition $a_1 = 2$.

$\mathcal{A}$. **Forward Substitution Steps:**

$$\begin{aligned}
a_2 &= 2 + 3 \\
a_3 &= (2 + 3) + 3 = 2 + 3 \cdot 2 \\
a_4 &= (2 + 3 \cdot 2) + 3 = 2 + 3 \cdot 3 \\
&\vdots \\
a_n &= a_{n-1} + 3 = 2 + 3(n-1)
\end{aligned}$$

So one of the closed-form formula $a_n = 2 + 3(n-1)$.

Iterative Solution: Forward Substitution

$\mathcal{Q}$. Let $a_n = a_{n-1} + 3$ for $n \geq 2$, with initial condition $a_1 = 2$.

$\mathcal{A}$. **Forward Substitution Steps:**

$$
\begin{aligned}
a_2 &= 2 + 3 \\
a_3 &= (2+3) + 3 = 2 + 3 \cdot 2 \\
a_4 &= (2 + 3 \cdot 2) + 3 = 2 + 3 \cdot 3 \\
&\vdots \\
a_n &= a_{n-1} + 3 = 2 + 3(n-1)
\end{aligned}
$$

So <u>one of the</u> closed-form formula $a_n = 2 + 3(n-1)$.

Iterative Solution: Backward Substitution

$\mathcal{Q}$. Let $a_n = a_{n-1} + 3$ for $n \geq 2$, with initial condition $a_1 = 2$.

$\mathcal{A}$. **Backward Substitution Steps:**

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\ &= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \\ &\vdots \\ &= a_2 + 3(n-2) \\ &= (a_1 + 3) + 3(n-2) = 2 + 3(n-1) \end{aligned}$$

Both iterative paths yield the same closed formula!

Iterative Solution: Backward Substitution

$\mathcal{Q}$. Let $a_n = a_{n-1} + 3$ for $n \geq 2$, with initial condition $a_1 = 2$.

$\mathcal{A}$. **Backward Substitution Steps:**

$$
\begin{aligned}
a_n &= a_{n-1} + 3 \\
&= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\
&= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \\
&\vdots \\
&= a_2 + 3(n-2) \\
&= (a_1 + 3) + 3(n-2) = 2 + 3(n-1)
\end{aligned}
$$

Both iterative paths yield the same closed formula!

Quick Check: Solving a Recurrence Relation

$\mathcal{Q}$. Solve the recurrence relation

$$a_n = a_{n-1} + 2n + 3, \qquad a_0 = 4.$$

Quick Check: Solving a Recurrence Relation

$\mathcal{Q}$. Solve the recurrence relation

$$a_n = a_{n-1} + 2n + 3, \qquad a_0 = 4.$$

$\mathcal{A}$. **Forward substitution:**

$$\begin{aligned} a_1 &= a_0 + 2(1) + 3 = 4 + 5 = 9, \\ a_2 &= a_1 + 2(2) + 3 = 9 + 7 = 16, \\ a_3 &= a_2 + 2(3) + 3 = 16 + 9 = 25, \\ a_4 &= a_3 + 2(4) + 3 = 25 + 11 = 36. \end{aligned}$$

Thus, the sequence begins

$$4, 9, 16, 25, 36, \ldots$$

Quick Check: Solving a Recurrence Relation

$\mathcal{Q}$. Solve the recurrence relation

$$a_n = a_{n-1} + 2n + 3, \qquad a_0 = 4.$$

$\mathcal{A}$. **Forward substitution:**

$$\begin{aligned}
a_1 &= a_0 + 2(1) + 3 = 4 + 5 = 9, \\
a_2 &= a_1 + 2(2) + 3 = 9 + 7 = 16, \\
a_3 &= a_2 + 2(3) + 3 = 16 + 9 = 25, \\
a_4 &= a_3 + 2(4) + 3 = 25 + 11 = 36.
\end{aligned}$$

Thus, the sequence begins

$$4, 9, 16, 25, 36, \ldots$$

$\mathcal{A}$. The terms are perfect squares:

$$\boxed{a_n = (n+2)^2}.$$

Quick Check: Solving a Recurrence Relation

$\mathcal{Q}$. Solve the recurrence relation: $a_n = na_{n-1}, \qquad a_0 = 5.$

Quick Check: Solving a Recurrence Relation

$\mathcal{Q}$. Solve the recurrence relation: $a_n = na_{n-1}, \qquad a_0 = 5.$

$\mathcal{A}$. The recurrence says: **To get the next term, multiply the previous term by n.** So the multipliers as we move through the sequence are

$$1,\ 2,\ 3,\ 4,\ 5, \ldots$$

Starting from $a_0 = 5$, we have

$$a_n = 5 \times 1 \times 2 \times 3 \times \cdots \times n. = \boxed{a_n = 5n!}.$$

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

$\mathcal{A}$. A growth rate of 1.12% means that each year the population increases by $0.0112a_{n-1}$. Therefore, $a_n = a_{n-1} + 0.0112a_{n-1} = 1.0112a_{n-1}$.

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

$\mathcal{A}$. A growth rate of 1.12% means that each year the population increases by $0.0112a_{n-1}$. Therefore, $a_n = a_{n-1} + 0.0112a_{n-1} = 1.0112a_{n-1}$.

$\mathcal{Q}$. Find an explicit formula for a_n.

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

$\mathcal{A}$. A growth rate of 1.12% means that each year the population increases by $0.0112a_{n-1}$. Therefore, $a_n = a_{n-1} + 0.0112a_{n-1} = 1.0112a_{n-1}$.

$\mathcal{Q}$. Find an explicit formula for a_n.

$\mathcal{A}$. Each year we multiply the population by the same factor 1.0112. Therefore, this is a geometric progression. Hence,

$$\boxed{a_n = 7.6(1.0112)^n}.$$

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

$\mathcal{A}$. A growth rate of 1.12% means that each year the population increases by $0.0112a_{n-1}$. Therefore, $a_n = a_{n-1} + 0.0112a_{n-1} = 1.0112a_{n-1}$.

$\mathcal{Q}$. Find an explicit formula for a_n.

$\mathcal{A}$. Each year we multiply the population by the same factor 1.0112. Therefore, this is a geometric progression. Hence,

$$\boxed{a_n = 7.6(1.0112)^n}.$$

$\mathcal{Q}$. What will the population be in 2050?

Quick Check: Modeling Population Growth

$\mathcal{Q}$. Assume that the world population in 2017 was 7.6 billion and is growing at a rate of 1.12% per year. Let a_n denote the population n years after 2017. Set up a recurrence relation for a_n.

$\mathcal{A}$. A growth rate of 1.12% means that each year the population increases by $0.0112a_{n-1}$. Therefore, $a_n = a_{n-1} + 0.0112a_{n-1} = 1.0112a_{n-1}$.

$\mathcal{Q}$. Find an explicit formula for a_n.

$\mathcal{A}$. Each year we multiply the population by the same factor 1.0112. Therefore, this is a geometric progression. Hence,

$$\boxed{a_n = 7.6(1.0112)^n}.$$

$\mathcal{Q}$. What will the population be in 2050?

$\mathcal{A}$. The year 2050 is $2050 - 2017 = 33$ years after 2017. Therefore,

$$a_{33} = 7.6(1.0112)^{33}.$$

Thank You for your attention!

Any questions?